# MATRIX PRICING

## Learning Outcome

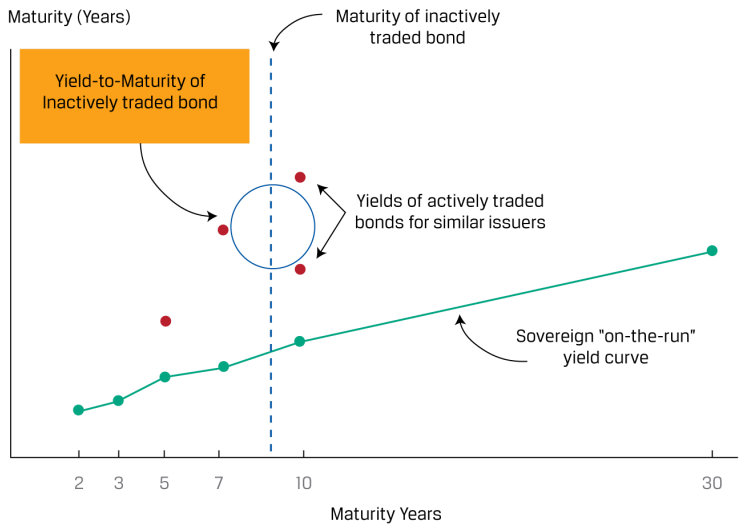
- describe matrix pricing

## Matrix Pricing Process

Unlike listed equity securities, most bonds are not actively traded, so there is often no current market price available to calculate yield-to-maturity. The same problem occurs for bonds that are not yet issued. In these situations, it is common to estimate price by using the prices of comparable but more frequently traded bonds. These comparable bonds have similar times-to-maturity, coupon rates, and credit quality. This estimation process, illustrated in Exhibit 8 and Exhibit 9, is called **matrix pricing**.

### Exhibit 8:

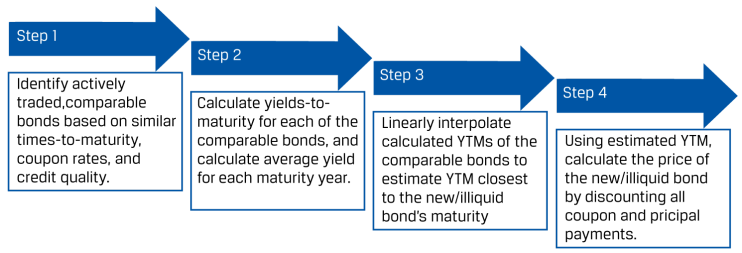
### Matrix Pricing




▶ Expand / collapse Extended Description

### Exhibit 9:

### Determining Price of New or Illiquid Bond Using Matrix Pricing

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there are no recent transactions reported for this particular security. Exhibit 10 and Exhibit 11 shows quoted prices and yields-to-maturity for four corporate bonds with very similar credit quality, in tabular and matrix form (by coupon rate and time-to-maturity).

### Exhibit 10: Actively Traded Comparable Bonds

| Bond | Tenor | Coupon (S/A) | Price | Yield-to-Maturity |
|---|---|---|---|---|
| A | 2 years | 3.00% | 98.5 | 3.786% |
| B | 2 years | 5.00% | 102.25 | 3.821% |
| C | 5 years | 2.00% | 90.25 | 4.181% |
| D | 5 years | 4.00% | 99.125 | 4.196% |

▶ Expand / collapse Extended Description

### Exhibit 11: Actively Traded Comparable Bonds Matrix

| | 2% Coupon | 3% Coupon | 4% Coupon | 5% Coupon |
|---|---|---|---|---|
| Two Years | | 98.500<br>3.786% | | 102.250<br>3.821% |
| Three Years | | | Bond X | |
| Four Years | | | | |
| Five Years | 90.250<br>4.181% | | 99.125<br>4.196% | |

▶ Expand / collapse Extended Description

Next, the analyst calculates the average yield for each year: 3.8035% for the two-year bonds and 4.1885% for the five-year bonds.

$$\frac{0.03786 + 0.03821}{2} = 3.8035\%$$

$$\frac{0.04181 + 0.04196}{2} = 4.1855\%$$

The estimated three-year yield for bonds of similar credit quality can be obtained with linear interpolation. The interpolated yield is 3.932%.

$$0.038035 + \left(\frac{3-2}{5-2}\right) \times (0.041885 - 0.038035) = 0.03932.$$

Using 3.9318% as the estimated three-year annual yield, the three-year, 4% semiannual coupon payment corporate bond has an estimated price of 100.191 per 100 of par value.

$$\frac{2}{(1.019659)^1} + \frac{2}{(1.019659)^2} + \frac{2}{(1.019659)^3} + \frac{2}{(1.019659)^4} + \frac{2}{(1.019659)^5} + \frac{102}{(1.019659)^6} = 100.191.$$

Calculation of yield-to-maturity can also be done with the Excel or Google Sheets YIELD function:

=YIELD(DATE(2025,10,1),DATE(2028,10,1),0.04,100.191,100,2,1)

Matrix pricing also is used in underwriting new bonds to get an estimate of the **required yield spread** over the benchmark rate, or the difference in yield-to-maturity between the bond and that of a government benchmark bond with the same or a similar time-to-maturity.

is 102.400 per 100 of par value. This is the full price, which is the same as the flat price because the accrued interest is zero. This implies that the coupon payment has just been made and there are four full years to maturity. The four-year rate of return required by investors for this bond is 2.36%:

$$102.400 = \frac{3}{(1+r)^1} + \frac{3}{(1+r)^2} + \frac{3}{(1+r)^3} + \frac{103}{(1+r)^4}; r = 0.0236.$$

This can also be calculated using the YIELD function in Microsoft Excel or Google Sheets:

=YIELD(DATE(2025,10,1),DATE(2029,10,1),0.03,102.401,100,2,1)

Suppose that there are no four-year government bonds to calculate the required yield spread on this security. However, there are three-year and five-year government bonds that have yields-to-maturity of 0.75% and 1.45%, respectively. The average of the two yields-to-maturity is 1.10%, which is the estimated yield for the four-year government bond. Therefore, the estimated required yield spread is 126 bps over the implied benchmark rate (0.0236 – 0.0110 = 0.0126).

The issuer now has an estimate of the four-year yield spread: 126 bps. This spread is a reference point for estimating the five-year spread for the newly issued bond.

## Question Set

**1.** True or false: Bond price quotes on such platforms as Bloomberg often represent the price at which the most recent trade for that bond was made.

**A.** True
**B.** False

Hide Solution

**Solution:**

False. Most bonds are infrequently traded, so price quotes on such platforms as Bloomberg are often estimated using matrix pricing methods.

**2.** An analyst is estimating a price for an illiquid four-year, 4.5% annual coupon payment corporate bond. The analyst identifies two corporate bonds that have similar credit quality: One is a three-year, 5.50% annual coupon payment bond priced at 107.500 per 100 of par value, and the other is a five-year, 4.50% annual coupon payment bond priced at 104.750 per 100 of par value. Using matrix pricing, the estimated price of the illiquid bond per 100 of par value is *closest* to:

**A.** 103.895.
**B.** 104.991.
**C.** 106.125.

Hide Solution

**Solution:**

B is correct. The first step is to determine the yields-to-maturity on the observed bonds. The required yield on the three-year, 5.50% bond priced at 107.500 is 2.856%:

$$107.500 = \frac{5.50}{(1+r)^1} + \frac{5.50}{(1+r)^2} + \frac{105.50}{(1+r)^3}; r = 0.02856.$$

The required yield on the five-year, 4.50% bond priced at 104.750 is 3.449%:

$$104.750 = \frac{4.50}{(1+r)^1} + \frac{4.50}{(1+r)^2} + \frac{4.50}{(1+r)^3} + \frac{4.50}{(1+r)^4} + \frac{104.50}{(1+r)^5}; r = 0.03449.$$

$$\frac{0.02856 + 0.03449}{2} = 0.031525.$$

Given an estimated yield-to-maturity of 3.153%, the estimated price of the illiquid four-year, 4.50% annual coupon payment corporate bond is 104.991 per 100 of par value:

$$\frac{4.50}{(1.031525)^1} + \frac{4.50}{(1.031525)^2} + \frac{4.50}{(1.031525)^3} + \frac{104.50}{(1.031525)^4} = 104.991.$$

**3.** An analyst uses the matrix pricing approach to determine the price of an illiquid bond. Identify three bond features that could be used to compare frequently traded bonds with the illiquid one.

Hide Solution

**Solution:**

Matrix pricing involves determining the price of a new or illiquid bond by comparing it to frequently traded bonds with similar

1. maturity,
2. coupon rates, and
3. credit quality.

**OPTIONAL UNTESTED SEGMENT**

## UNLOCKING REAL-WORLD APPLICATIONS WITH BLOOMBERG

Bloomberg is one of the main market data platforms used globally by finance professionals throughout their workflows. In this optional section, we will explore a real-world application of the CFA Curriculum on Bloomberg.

### {PT} PRICE TO YIELD TABLE

The Bloomberg PT function allows you to view a chart for any bond plotting price against yield.

> "PT provides a table and graph of yields and prices for a selected government, corporate security, or preferred security, so you can quickly compare yields for a range of prices. PT allows you to quickly convert bond yields to maturity into prices and vice-versa." -Bloomberg

To explore the PT function, let's first identify a bond to research. Type SECF into the Bloomberg command line, and press Enter or <GO>. In the top left amber Category dropdown menu, select Fixed Income. Click on one of the bonds listed. Then type PT into the Bloomberg command line, and press Enter or <GO>. PT is divided into a control area and a series of data columns that illustrate the relationship between price and yield. The Table and Chart buttons on the right allow you to switch between a table and chart view of yield/price data.

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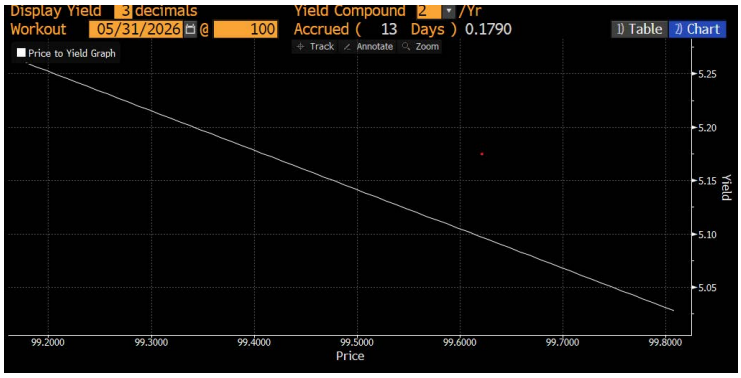


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LB Where are the 3-2 and 5-2 derived from in the linear interpolation?

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